Chest Pain Risk — HEART Score & T-MACS

adults with suspected ACS, non-diagnostic ECG · casualty

HEART 0–10

HEART SCORE

	0	1	2
History	Slightly suspicious	Moderately suspicious	Highly suspicious
ECG	Normal	Non-specific repolarisation (LBBB, LVH, paced, digoxin)	Significant ST deviation
Age	<45 y	45–64 y	≥65 y
Risk factors*	None known	1–2	≥3 or known atherosclerotic disease
Troponin	≤ normal limit	1–3 × normal limit	>3 × normal limit

HEART — 6-WEEK MACE

- **0–3** Low (≈1.7%) — consider discharge after serial troponin
- **4–6** Moderate (≈12–17%) — admit, serial troponin, cardiology
- **7–10** High (≈50–65%) — early invasive strategy

NOTES

- *Risk factors: hypertension, hypercholesterolaemia, DM, obesity (BMI >30), smoking (current or quit ≤3 months), family history
- Do not use if ST elevation (STEMI pathway) or haemodynamically unstable

T-MACS VARIABLES (ON ARRIVAL)

- ECG ischaemia
- Worsening or crescendo angina
- Pain radiating to the right arm or shoulder
- Pain associated with vomiting
- Sweating observed
- Hypotension (SBP <100 mmHg)
- hs-troponin T (Roche, ng/L) on arrival — single sample

T-MACS PROBABILITY OF ACS

- **<2%** Very low — ACS ruled out; consider discharge
- **2–5%** Low — repeat troponin at 3 h, low-dependency area
- **5–95%** Moderate — serial troponin, admit to ward
- **≥95%** High — ACS ruled in; cardiology

T-MACS NOTE

- Logistic equation validated for Roche hs-cTnT only — use the official calculator; coefficients not printed here (not verified)
- Rules out ≈40% with one blood test (sensitivity ≈98%, NPV ≈99%)

SA STG

- Repeat ECG at 20–30-min intervals if the initial ECG is not diagnostic ✚
- Do not delay treatment while waiting for troponin results ✚

HOSP STG & EML Hospital Level Adults, 6th ed. 2024 (updated Mar 2026) — pp 3.4 · **HEART** Six AJ et al. Neth Heart J 2008;16:191-6; Backus BE et al. Int J Cardiol 2013;168:2153-8 · **TMACS** Body R et al. T-MACS decision aid. Emerg Med J 2017;34:349-56

✚ = SA STG / national guideline (pages listed above). Unmarked content: international source named above. ‡ = STG misprint or sources differ (see app). ANDH · chest-pain · updated 2026-10-14